

Soham Khataavkar

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Education

Indian Institute of Technology Bombay

July 2025 – Present

M.Tech in Computer Science – CGPA: 9.21/10.0

Mumbai, Maharashtra

National Institute of Technology Nagpur

July 2021 – May 2025

B.Tech in Computer Science and Engineering – CGPA: 8.32/10.0

Nagpur, Maharashtra

Projects

sIDE+ | MongoDB, ExpressJS, ReactJS, NodeJS, Azure, Docker, Socket.IO, BullMQ Demo | Frontend | Backend

- Developed a full-featured **online code execution platform** supporting C, C++, Java, Python, and JavaScript.
- Integrated **Docker-based isolated execution environments** using Docker images (e.g., **GCC**, **OpenJDK**) with features like **execution time limits** and **automatic container cleanup**, to enhance security.
- Implemented features such as **auto-save**, **auto-completion**, **cloud-save**, and **code-share** within a rich code editor interface integrated with **Gemini AI** to enhance the user experience.

CrHash | C, C++, CUDA, Make

GitHub

- Developed a **hardware-accelerated** password cracking system for MD5 and SHA-1 using **CUDA** and **C++** to evaluate billions of hashes concurrently by maximizing thread-level parallelism on the GPU.
- Implemented **brute-force** and **dictionary attack** with batched input processing to maximize GPU utilization.
- Achieved a **78.2x** speedup, sustaining a peak throughput of **5.6B hashes/sec** for MD5, and a **61.4x** speedup reaching **4.05B hashes/sec** for SHA-1 over optimized **multi-threaded** CPU baselines.

CuSync | C++, CUDA, OpenCV, CMake

GitHub

- Developed a **high-performance video processing engine** in C++ and CUDA, implementing **custom CUDA image processing kernels** for optimized memory access patterns and thread block execution.
- Built an **asynchronous execution pipeline** via CUDA streams to overlap PCIe data transfers with GPU computation, significantly reducing processing latency and eliminating wasted hardware cycles.
- Integrated an interactive OpenCV HighGUI interface for real-time playback controls and live parameter tuning.

Setris | C++, Raylib, Make

Demo | GitHub

- Developed a custom implementation of **Tetris** in C++, enabling players to fully customize gameplay rules such as speed, block shapes, and drop behavior.
- Built with a fully modular architecture leveraging core **Object-Oriented Programming (OOP)** principles, enabling clean separation of concerns, enhanced code reusability, and ease of scalability and maintenance.

Achievements

- Winner – Microland Hackathon**; built a full-stack AI-powered health chatbot, leading to a **full-time Software Developer** offer from Microland.
- Selected as Deputy Field Officer (Technical)** at the **Cabinet Secretariat**.
- Selected as Junior Executive (IT) Engineer** at the **Airports Authority of India**.
- Selected as Executive Engineer** at the **National Thermal Power Corporation Limited**.

Technical Skills

Languages: C, C++, CUDA, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: ReactJS, NodeJS, ExpressJS, Raylib, REST APIs, BullMQ, Socket.IO

Databases & Cloud: MongoDB, MySQL, Azure, Cloudinary, Render

Tools & Build Systems: Docker, CMake, Make, Git, GitHub, CUDA Toolkit, Linux

Relevant Coursework

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|-------------------|---------------------|---------------------|-----------------------|
| • Data Structures | • Operating Systems | • Computer Networks | • Database Management |
| • OOP | • Algorithms | • Web Security | • GP-GPU |